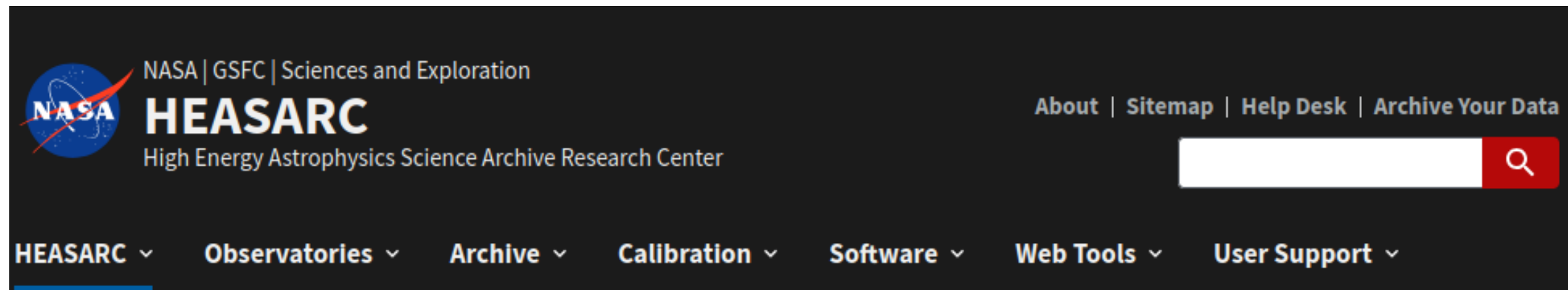


HEASARC | HEASoft



NASA's designated multi-mission astronomy archive for cosmic astronomy datasets for observations in the extreme ultraviolet (EUV), **X-ray**, and gamma-ray wave bands, as well as for cosmic microwave background (CMB) datasets.

<https://heasarc.gsfc.nasa.gov/>

Active Missions

AstroSat

BurstCube

CALET

Chandra

Fermi

HXMT-Insight

Integral

IXPE

MAXI

NICER

NuSTAR

SRG eRosita /
ART-XC

Swift

XMM-Newton

XRISM

Past Missions

AGILE

ANS

ARIEL V

ASCA

BBXRT

BeppoSAX

CGRO

COPERNICUS

COS-B

DXS

EINSTEIN

EUVE

EXOSAT

GINGA

GRANAT

HaloSat

HAKUCHO

HEAO-1

HEAO-3

HETE-2

Hitomi

OSO-7

OSO-8

ROSAT

Rossi XTE

SAS-2

SAS-3

Suzaku

TENMA

UHURU

VELA 5B

WASS

XQC

Archive

XAMIN: The primary graphical user interface portal for exploring the data holdings, including catalogs and data products.

<https://heasarc.gsfc.nasa.gov/xamin/>

astroquery.heasarc: HEASARC maintains an astroquery module for uniform access to world-wide astronomy data following the Astropy model.

<https://astroquery.readthedocs.io/en/latest/heasarc/heasarc.html>

HEASARC Browse: Online tool from NASA that allows users to search and access astronomical data from high-energy astrophysics missions and catalogs.

<https://heasarc.gsfc.nasa.gov/cgi-bin/W3Browse/>

Skyview: A powerful tool to visualize the sky with the largest collection of multiwavelength sky maps from the radio to the gamma-ray.

<http://skyview.gsfc.nasa.gov/>

Cloud & Platforms



HEASARC@SciServer is an analysis-ready platform where both data and software are available through the browser.

Software

HEASoft: Unified package of several High Energy tools including:

Cfitsio: A library for reading and writing FITS files for C and Fortran programmers.

Ftools: General FITS file utility programs and mission-specific data analysis tools.

XANADU: Suite of spectral (xspec), timing (xronos), and image (ximage) analysis programs.

Xselect: Multipurpose tool for filtering event files and generating images, spectra, and light curves.

Ftools: General FITS file utility programs and mission-specific data analysis tools.

HEASoftpy: A Python 3 interface to HEASoft.

FV: Interactive editor and viewer for astronomical data files in FITS format.

XSTAR: Program for calculating physical conditions and emission spectra in photoionized gases.

<https://heasarc.gsfc.nasa.gov/docs/software/lheasoft/>

FITS Data Format

FITS stands for '**Flexible Image Transport System**' and is the standard astronomical data format endorsed by both NASA and the IAU.

FITS is much more than an image format (such as JPG or GIF) and is primarily designed to store scientific data sets consisting of multi-dimensional arrays (1-D spectra, 2-D images or 3-D data cubes) and 2-dimensional tables containing rows and columns of data.

